

# User Stories and UX Design Requirements

## Highlights

- The concept of requirements for UX design
- User stories as the agile way to express UX design requirements
  - Definition of a user story
  - The user story structure
  - Writing user stories
- UX design requirements are needed in rare situations
  - Composing requirement statements
- Validating user stories and requirements with users and stakeholders

## 10.1 INTRODUCTION

### 10.1.1 You Are Here

We begin each process chapter with a “You are here” picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 10.1).

The aim of UX research (all of Part 2) is to discover and learn about wants and needs for the UX design. In this chapter, we synthesize information about those wants and needs into user stories and design requirements to drive the design of the target product or system.

### 10.1.2 The Difficult Reality of User Stories and Requirements

For clarity in the previous edition of this book, we attempted to discuss user stories and requirements independently of other UX design issues. We hoped this would help the reader understand the basic concepts, but it also concealed the complexity of their reality. User stories and requirements do not come into being at the end of UX research and neatly define what needs to be designed next. Some of them are evident from project conception, and others are discovered late in the development process in response to new insights gained

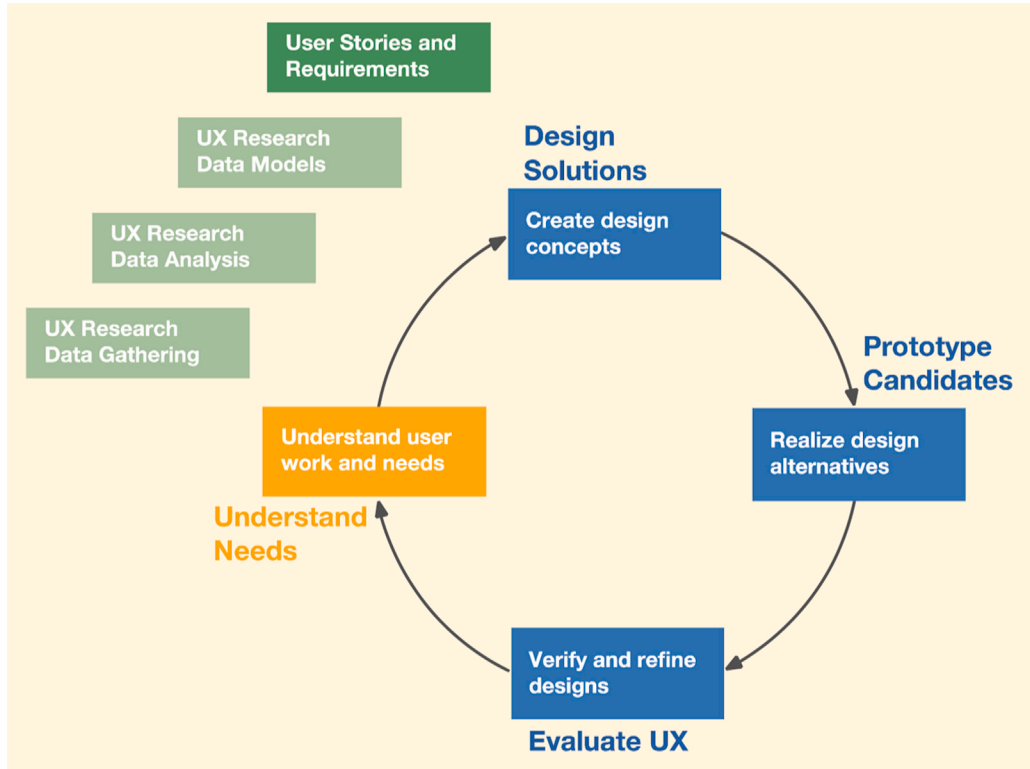


Fig. 10.1

*You are here in the user stories and UX requirements chapter within the understand needs activity, in the context of the overall Wheel UX lifecycle.*

as the team becomes immersed in design or as part of evaluation activities (see Part 5). It is not uncommon for the project teams to encounter technical constraints after software developers attempt to realize design specifications in code, forcing a change or even additions to requirements.

Even in the idealized flow of UX research activities as laid out in this book thus far, and leading up to this chapter, there is no single process flow to drive the requirements activity.

Everything we do in the UX research of Part 2 funnels into the user stories and requirements of this chapter. It is a culmination of the synthesis activity, where everything we learned about the work<sup>1</sup> practice and its domain are now used as inputs to define the envisioned solution and how it can improve that practice. What happens here plays a key role in how well the UX research process serves UX design in your project.

<sup>1</sup>The term work broadly signifies anything people/users do while using products or systems (see Section 6.2).

### 10.1.3 Completeness in Requirements Is Not Feasible

In the traditional waterfall approaches to building systems, there was a heavy emphasis on completeness of requirements. Specialized roles would spend much time and effort on detailing every aspect of the envisioned system before attempting to begin the design activity. In practice, this rarely worked because of the reality that the team's understanding, capabilities of the technology, customer's needs, or myriad other things would inevitably change between the time the requirements were specified and the system was built.

In response, the agile approaches embraced a less complete, and rather informal, approach, where we pose requirements for the envisioned system as user stories and, sometimes, as general requirements. More on these user stories is coming up in the next section.

### 10.1.4 Getting From Models and the Work Activity Affinity Diagrams to User Stories and Requirements

One of the easiest ways to begin looking for requirements is from the notes in the work activity affinity diagram (WAAD); however, because these are almost always incomplete, they cannot be the only source for requirements. We also need to include the more complete data models as inputs. For example, each task in a task inventory needs to be covered in the requirements. So must every sequence in the task sequence model and every flow in the flow model. Each breakdown in these models must be addressed in the requirements as well.

Following is an overview to get you started. In this chapter, we will derive requirements for the system we are designing from the data models and other inputs from customers and users.

- Go through each node (work activity note) of the WAAD and, if warranted, derive user stories and/or general requirements represented by it.
- Go through each data model and, if warranted, extract corresponding user stories and/or general requirements.
- Use the close collaboration with customers that agile contexts provide to gather inputs and priorities for the emerging requirements.
- More generally, be aggressive in seeking out as many requirements from other inputs as possible.

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## 10.2 USER STORIES

In today's world, user stories are the standard approach to both UX design requirements and software engineering (SE) implementation requirements in late-funnel (see [Section 4.4.3](#)) agile development.

### 10.2.1 What Is a User Story?

A user story is a small-scope UX requirement written as a short narrative, expressed in the user's voice, describing a feature or capability needed by a user in a specific work role and a rationale for why it is needed, used as an agile UX design requirement.

According to Gothelf and Seiden (2016), a user story is a kind of very small-scope (see [Section 4.3.2](#)) requirement that designates “the smallest unit of work expressed as a benefit to the end user.”

In the classic agile software literature, customers write these user stories and assign priority ranking to each. These stories are then fed into the development pipeline, defining the scope for each late-funnel release. As such, they indicate what is most valuable for the customer and act as a prioritization mechanism.

User stories:

- Involve user inputs
- Are stated from a user and usage perspective
- Were made popular by agile software proponents
- Are now the de facto standard in industry
- Are the means for delivering chunks of meaningful capabilities to users

#### 10.2.1.1 *Stories exclusively from users cannot produce complete requirements*

User stories derive from raw UX research data in which users articulate their needs and desires for functionality and UX features. Early on, though, it became clear that users, not trained to produce requirements, are unlikely to present us with a complete set of the same. We get what we can from users, but inputs to user stories must come from many sources (e.g., subject-matter experts, work domain or industry regulations, marketing needs). Wherever inputs originate, agile approaches emphasize close communication with, and strong emphasis on, users and their involvement in the development process. The goal is for the development team to deliver small-scope releases that bring most business value to the customer. Requirements and capabilities that originate from other sources can still be validated and prioritized by users.

#### 10.2.1.2 *Clean up and augment user stories*

At the end of the day, it is up to the UX analyst to ensure the quality and relative completeness of user stories used to drive UX design. We, as UX analysts, must:

- Edit existing user stories to make them effective
- Edit inputs from the many sources into the format of user stories

- Write additional user stories to fill in gaps
- Cover in user stories all requirements handed down in contracts and project definition documents

Irrespective of who wrote them, user stories must capture design needs and divide up requirements into items of small scope.

## Example: Identifying Inputs to User Stories for the Ticket Kiosk System

Consider these related work activity notes:

*I like to keep up to date with what is going on regarding entertainment in the community.*

*I especially want to know about current and popular events.*

Each of these sentences can be interpreted as an input to a user story.

## Example: Identifying inputs to user stories for the Ticket Kiosk System

Consider these two related work activity notes:

*I want to see reviews and other feedback from people who have already seen the show.*

*[Design idea] Consider including capability for people to add reviews and to rate reviews.*

The first is an obvious candidate as an input to a user story. The second, added by the UX research data gatherer as a related thought, is a design idea that may well be transformed into the form of a user story. Although technically this design idea did not come directly from a user, it did arise in the context of a user interview and, if valuable to the process, should be captured.

### 10.2.2 Writing User Stories

#### 10.2.2.1 Approximate format for a user story

Beginning with inputs from the customer, user, or suitable proxy, identify the highest priority areas of the envisioned system to focus on. Look at the work activity notes, WAAD, and other models you have synthesized that target those areas. Look for insights reflecting problems users have with an existing product or system as well as needs for new features (e.g., breakdowns in the flow model or key tasks in the hierarchical task inventory [HTI]). Now is the time to convert them into user stories if you have not already progressed to where you write them directly from raw UX research data or even in data gathering (see [Chapter 6](#)).

Because of its small size and scope, each user story should fit on one 3×5 index card. We use the following format to represent a user story:

As a <relevant user work role>  
 I want to be able to <the desired small-scope capability in product or system usage>  
 So that <the reason why, the added value this capability will deliver>.

## Example: User Story for the Ticket Kiosk System

Consider this work activity note:

*For sporting events I want the option to buy several tickets with adjacent seating.*

The next step to a user story is a short one:

As a student ticket buyer  
 When buying tickets for sporting events, I want the option to get several seats together  
 So I can sit with my friends.

This is a small-scope statement referring to one choice in seat selection as a parameter of the ticket-buying task.

As we can see from these examples, it is often easy to write a user story directly from the raw data note or even from the data gathering observations and interviews, while skipping the steps of composing and extracting the associated work activity note.

## Examples: Ticket Kiosk System User Stories Directly From Raw UX Research Data

Consider this raw data note:

*I sometimes want to find events that have to do with my own personal interests.  
 For example, I really like ice skating and want to see what kinds of entertainment events in the nearby areas feature skating of any kind.*

This note is specific to ice skating, but you can deduce a slightly more general version for your user story:

As a ticket buyer  
 I want to be able to search for events by event type or descriptive keyword (e.g., ice skating)  
 So I can find the kinds of entertainment that I like.

Direct translation to a user story reduces steps, saves time, and is more responsive to an agile process.

Following is another example, beginning with this raw data note:

*I occasionally find it convenient to see recommendations for events that are similar to ones I like, like in the listing of what others who looked at this item got from Amazon.com.*

This note can be interpreted directly into a user story for a specific feature:

As a ticket buyer  
I want to be able to see recommendations for events similar to ones I find and like  
So I can find other events I like that I might otherwise miss.

This user story needs an attached note to describe an implied backend system requirement for the SE and systems people. The Ticket Kiosk System functional software may have to maintain user profiles to track the kinds of choices made by each ticket buyer. This similar capability at Amazon is a design model for this feature.

### 10.2.3 Extrapolation of Requirements in User Stories to Generalize UX Research Data

User statements in raw data notes, work activity notes, or UX research data models can be quite narrow and specific. Sometimes you will need to extrapolate from these statements or insights to get a more general and more useful case, as we did in the previous example about ice skating.

#### Example: Generalizing Ways to Find Events in the Ticket Kiosk System

Suppose potential ticket buyers expressed the need to search for events, but they said nothing about browsing events to see what is available. You may, then, want to create an additional user story as an extrapolation:

As a ticket buyer  
I want to be able to browse events with filters for category, description, location, time, rating, and price  
So I can find relevant events in context.

This user story will be grouped with any other user stories about searching, browsing, or finding tasks.

#### Example: Broadening a Requirement for Exchanging Tickets

As another example, in a work activity note, a ticket buyer says:

*I really would like to be able to post an MU football ticket for exchange with a ticket in another location in the stadium to be able to sit with my friends.*

In our extrapolated user story, we broadened this to:

As a ticket buyer  
I want to be able to post, check status of, and exchange student tickets  
So I can swap reserved tickets for seats where I can sit with my friends.

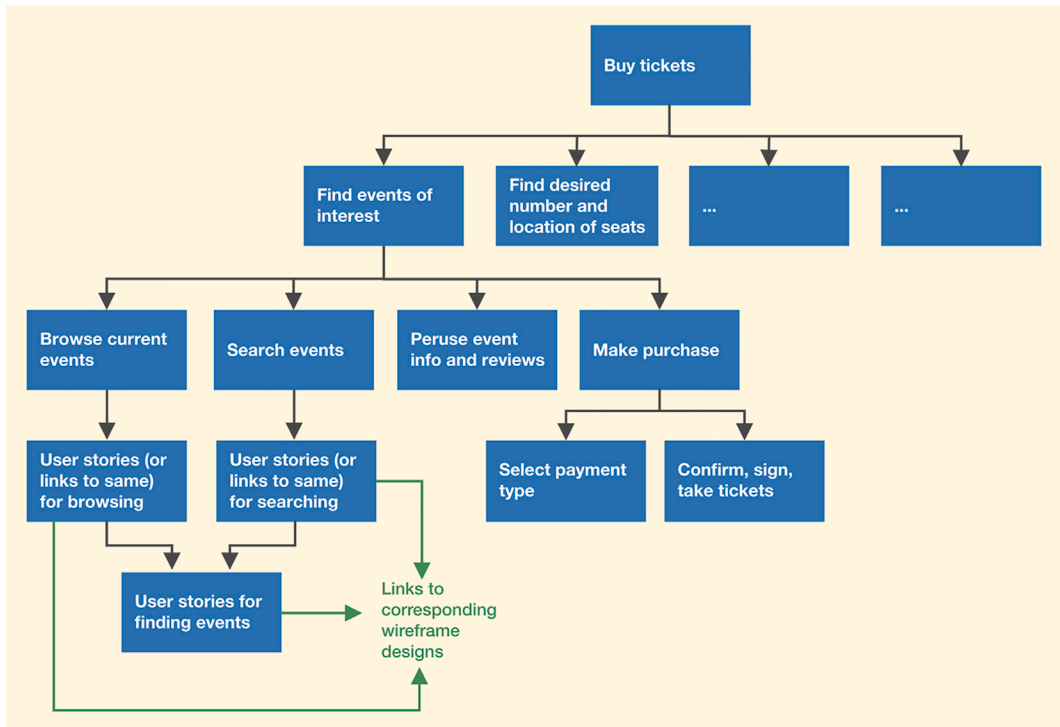
For SE and systems people, this user story also implies a corresponding system requirement for customer accounts of some kind, presumably where they can log in using their Middleburg University Passport IDs.

### 10.2.4 The User Story Structure: Organizing Sets of User Stories for Use in UX Design

In a project, you may need to organize a large number of diverse user stories by grouping sets of related features or related tasks in what we loosely call the user story structure. Because UX user stories are stories about user capabilities, they are closely related to user tasks, and you can use structures similar to the HTI (see [Section 9.2](#)) to organize them. This allows us to consider related features together. For example, we should look at the browse-by-event-type task near other browsing tasks, not far from searching tasks. Organizing around the HTI is a way to manage the collection of user stories. The need for developing the user stories structure is one reason we need the early part of the agile UX funnel model, the large-scope activity part of the funnel for modeling task structure and conceptual design.

#### Example: Grouping of User Stories Related to Finding and Buying Tickets for Events

Consider the partial task and user story structure for the ticket buyer work role, as shown in [Fig. 10.2](#).



*Fig. 10.2*  
A partial task and user story structure for the ticket buyer work role.



## Example: User Stories for Finding Events

The box at the lower left in [Fig. 10.2](#) contains a group of user stories that are related to the user tasks of browsing and searching to find an event of interest for which to buy tickets:

### Searching

#### User story 83:

As a ticket buyer  
I would like to be able to search for events by price, artist, venue, and/or date  
Because I want maximum flexibility in searching.

#### User story 104:

As a ticket buyer  
I expect a Google-style search where I type in, for example, the name of a band  
Because I don't want to be constrained by special formats in searches.

#### User story 112:

As a ticket buyer  
I want to be able to search for events by proximity to a given location  
Because I might want to find events near me.

### Browsing

#### User story 74:

As a ticket buyer  
I would like to be able to browse all events on the kiosk  
Because sometimes direct searches don't reveal enough context.

#### User story 106:

As a ticket buyer  
I would like to be able to browse by different event characteristics, such as music genre  
So I can find events I like when I only know the types of events I'm looking for.

### Sorting

#### User story 75:

As a ticket buyer  
I would like to be able to sort the results I find using some criteria  
Because sometimes searching produces too many results to go through without any ordering.

## Exercise 10.1 Write User Stories for Your Product or System

Using the format suggested in [Section 10.2.2.1](#), write a few plausible user stories for your product or system (see [Section 5.6.1.2](#)).

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## 10.3 UX DESIGN REQUIREMENTS

For completeness, we now wave a hand at a brief description of UX design requirement specification. As we said in [Section 4.2](#), back in the old days when the waterfall process was widespread, it was traditional to represent what is needed in the system in formal requirement documents, often articulated from a system-centered perspective.

However, even in the SE world, the importance of requirement specifications is fading, primarily because they do not work (Beck & Andres, 2004). There is an increasing recognition that detailed formal requirements cannot:

- Ever be complete
- Ever be 100% correct
- Be prevented from changing throughout the lifecycle

Thus formal requirements are almost never used in today's UX design. As a UX designer, it is also unlikely you will ever be asked to write requirement specifications. The rare exception may be for a very complex project or if you are contractually obligated to adopt formal requirements. This section will help you with that.

### 10.3.1 Requirements as Design Goals, Not Constraints

Before moving on to the mechanics of requirements, we offer another way to think about requirements, especially UX requirements, and that is as design goals. Requirements of yore were immutable and absolute: "You shall provide these features and functions, and you shall do it in the way we say." This dictatorial view of requirements left no room for creativity in design solutions and no flexibility to adapt to change as it inevitably occurred.

Today's UX mentality is all about valuing design over requirements (Kolko, 2015b). That is to say, rather than even thinking about requirements, we should think about how great products "require vision, perseverance, and the ability to do amazing things in the face of perceived impossibility" (Kolko, 2015b, p. 22).

In today's agile environments, we like to view requirements, especially UX design requirements, not as constraints but as design goals with the freedom to interpret how the design should achieve these goals (Kolko, 2015b).

### 10.3.2 Look at the WAAD and Other UX Research Data Models to Get Started With Requirements

We always look to the WAAD, built in [Section 7.5](#), for inputs to general requirements. Go through the WAAD one work activity note at a time. If any of the WAAD notes are best interpreted as inputs to requirements, then capture them as such.

Requirements can also come from all the other models. For example, each task in an HTI needs to be covered by a requirement. The same is true for a task sequence model; each task sequence must be supported in the envisioned system, making it a requirement.

### 10.3.3 The Requirement Statement

A suggested requirement statement format is shown in [Fig. 10.3](#), reflecting a want or need expressed in a work activity note or a model.

|                                                                             |
|-----------------------------------------------------------------------------|
| Name of major feature or category within requirements structure             |
| Name of second-level feature or category                                    |
| The requirement                                                             |
| Rationale (if useful): rationale statement (the reason for the requirement) |
| Note (optional): additional comments, if useful                             |

*Fig. 10.3*

*Generic structure of a requirement statement.*

### Example: Writing a Requirement Statement From a Ticket Kiosk System Work Activity Note

Consider this UX research work activity note from the Ticket Kiosk System:

*The user should get a printed ticket as part of a purchase transaction.*

Without this kind of immediate feedback, users can feel uncomfortable about having to trust that they will get what they paid for.

This note can directly morph into the high-level design requirement statement of [Fig. 10.4](#).

|                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ticket purchase sequence                                                                                                                                      |
| Tangible user feedback.                                                                                                                                       |
| <i>The kiosk shall produce printed tickets as part of a ticket purchase transaction.</i>                                                                      |
| <i>Note: This requirement has serious hardware and system implications. The ticket printer must be highly reliable and must receive periodic maintenance.</i> |

*Fig. 10.4*

*Example requirement statement about ticket purchase sequence.*

## Example: Writing a Ticket Kiosk System Elemental Data Note as a Requirement Statement

Consider this UX research data note from the TKS:

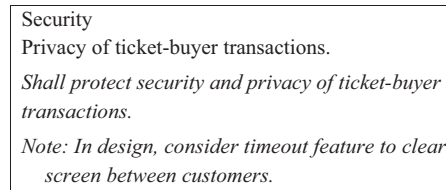
*I am concerned about the security and privacy of my transactions.*

Concern about security and privacy is a system-wide issue, so this note can directly lead to a high-level design requirement statement:

*Shall protect the security and privacy of ticket-buyer transactions.*

Note that at this level, requirements can involve both UX and functional requirements.

Finally, it is easy to fit this into the requirements structure, assuming there is already a category for personal privacy and trust issues (Fig. 10.5).



Security  
Privacy of ticket-buyer transactions.  
*Shall protect security and privacy of ticket-buyer transactions.*  
*Note: In design, consider timeout feature to clear screen between customers.*

Fig. 10.5

Example requirement statement about privacy and security.

## Example: Generalizing data notes to express Ticket Kiosk System needs as requirement statements

If the elemental data note is very specific, you may have to abstract it to express a general need. Consider the following raw data note:

*Normally, I buy tickets at the venue of the event just before the event. For special events, I might buy online. Then I usually have to buy Metro tickets to get there at a different kiosk in a totally different location. If I could buy all these different tickets at one kiosk, that would give me one-stop shopping for events.*

This is about treating events as broad entertainment activities and having sufficiently broad coverage of tickets for “event content” so the kiosk can be a one-stop shopping location for entertainment events. This means extending “event content” to include, for example, restaurant reservations and transportation tickets in conjunction with events (Fig. 10.6).

Business issues, decisions:  
 Kinds of events and tickets supported.  
*Shall provide tickets and reservations for a broad range of needs associated with an entertainment activity.*  
*Example: This should include tickets for events and transportation and restaurant reservations.*

Fig. 10.6

*Example requirement statement about kinds of supported events.*

## Example: A Possibly Unexpected Ticket Kiosk System Requirement

*Data note: Although marketing people might not think of putting a kiosk next to the ticket office, it would be very helpful for people who get there and find the ticket office too busy or closed.*

This data note (about locating a kiosk near a ticket office) may reveal an unexpected need because you may not think of it. A possible resulting requirement statement is shown in Fig. 10.7.

Business issues, decisions:  
 Kiosk locations.  
*Shall provide a kiosk located near the ticket office.*  
*Note: This requirement serves users when the ticket office is too busy or is closed.*

Fig. 10.7

*Example requirement statement about kiosk locations.*

### 10.3.4 Organizing Requirements Statements

You need to devise a way to manage and keep track of requirement statements. For projects not requiring maximum rigor, a simple indented list will suffice as an effective way to track system capabilities for which we will be designing. For larger projects and/or cases requiring more rigor, you can construct a requirements document, which is a hierarchical structure of requirement statements by category.

## Exercise 10.2 Write UX Design Requirement Statements for Your Product or System

Using the format suggested in Section 10.3.3, write a few plausible UX design requirement statements for your product or system (see Section 5.6.1.2).

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## 10.4 TAKE USER STORIES AND REQUIREMENTS BACK TO CUSTOMERS AND USERS FOR VALIDATION

After the user stories and/or requirements are gathered and organized into a hierarchical structure, it is a good time to take them back to users and other stakeholders to be sure they are the right ones. Your structured sets of user stories and requirements offer you, your customer or client, and your users a lens through which you can contemplate and discuss groups of related user stories and requirements at various levels of abstraction. This is an important step for you both because it gives you a chance to get input and to correct misconceptions before you get into design. It also helps solidify your relationship as partners in the process.

Walk them through the requirements to make sure your interpretation of user stories and requirements is accurate and relatively complete.

Because these requirements, by nature, go well beyond UX concerns and affect SE, hardware, and systems, it is a good time to coordinate SE and system engineering teams, raising awareness of requirements that affect their domains and fostering consistency of terminology and understanding.

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## 10.5 REVIEW PROGRESS TOWARD SYNTHESIZING AND UNDERSTANDING “THE ELEPHANT” THAT IS UX DESIGN NEEDS

The goal of UX research in the chapters of Part 2 is to achieve a thorough understanding of the design needs for a product or system to be designed. Understanding the design needs is a large and complex undertaking, depending on understanding the users and their work practice within the work domain.

You also need more than one perspective to get the full picture. Each UX research activity, such as data gathering, data distillation, WAAD building, data modeling, and requirements writing, reveals a different perspective for understanding design needs. It is much like the parable of the blind men trying to understand an elephant (source unknown), each feeling a different part. The blind men work together to combine each of their tactile perceptions to obtain a mental picture of the whole animal.

That is how we piece together what we learn from each UX research activity to understand the whole of design needs. The tool we use to integrate

the many different inputs is synthesis, a technique from the UX technique toolbox. In general, synthesis is used for combining diverse inputs, facts, and observations of parts of a concept to make conclusions that characterize and aid understanding of the whole concept (Kolko, 2010).

In UX research, we use synthesis to find connections among the models and representations we construct and put the pieces together to form the whole elephant. Here, we combine pieces obtained by the UX research data organized in different ways to arrive at a picture of the work practice and design needs for the product or system. We use synthesis techniques to tease out the connections through combination, fusion, and other ways of sifting through the data, unearthing hidden relationships and insights into the work practice and design needs in the product or system work domain. This is where immersion pays off.

Now that we have performed each of these UX research activities in Part 2 of this book, it is time to step back and review our progress toward synthesizing and understanding the elephant that is UX design needs. As a kind of litmus test, we submit the following questions:

- Do we have a complete understanding of the users, their goals, tasks, emotional impact factors, social dynamics; the work and work practice, work roles, including breakdowns, gaps, challenges, and opportunities?
- Do we understand the ecology of the work practice and the contrast with others in the domain?
- Do we understand the broader trends in the business domain?

If the answers are not all yes or a qualified yes, then designers need to flesh out that understanding with appropriate models or artifacts, such as WAADs or even user stories and requirements.

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## 10.6 SEGUE

Congratulations on getting through Part 2 on UX research. We now move into Part 3 of the book on design, followed by Part 4, which is about prototyping. The process chapters are completed in Part 5, about UX evaluation.